REVIEW

# Artificial Intelligence and Machine Learning (AI/ML) Revolution in Cardiology Medical Devices: From Diagnosis to Treatment

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**Abstract**

The integration of Artificial Intelligence (AI) and Machine Learning (ML) is revolutionizing cardiology, enhancing diagnostic precision, treatment efficacy, and overall patient care. AI-powered medical devices are transforming cardiovascular healthcare by enabling early disease detection, real-time monitoring, and personalized treatment strategies. Advanced imaging techniques, predictive analytics, and AI-driven wearable devices are improving clinical workflows and patient management. AI-based algorithms analyze vast datasets to identify subtle patterns in electrocardiograms, echocardiography, and cardiac MRI scans, leading to faster and more accurate diagnoses. Innovative devices such as the Medtronic Micra TPS, Abbott CardioMEMS™ HF System, and Boston Scientific Watchman Device exemplify the role of AI in optimizing cardiac interventions, reducing hospital readmissions, and improving long-term patient outcomes. AI-driven tools also facilitate remote monitoring and predictive risk assessment, ensuring timely medical interventions. These advancements are reshaping the landscape of cardiology by streamlining clinical decision-making, improving procedural efficiency, and expanding care accessibility. As AI/ML technologies continue to evolve, their integration into cardiology holds the potential to transform cardiovascular disease management, enhance preventive care, and enable precision medicine. Unlike prior reviews, this work provides a device-centered perspective by systematically highlighting AI/ML-enabled cardiology devices with regulatory approval, while also addressing safety, reliability, and ethical challenges to guide real-world clinical adoption.



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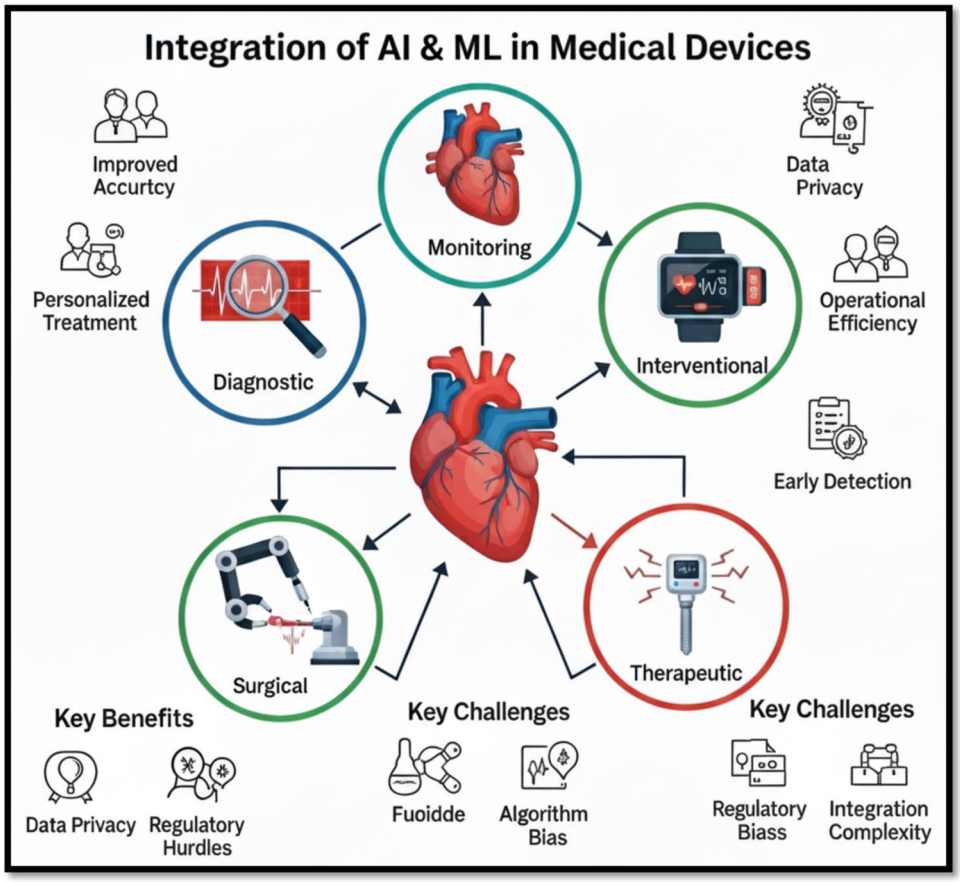
**Graphical Abstract**




## Highlights

- Innovative advancements in medical device technology and applications.
- Regulatory challenges and compliance in the medical device industry.
- Clinical implications and future trends in medical devices.
- Role of AI and IoT in healthcare innovation.

**Keywords** AI/ML · Cardiology medical devices · Diagnostic and therapeutic devices · FDA/CE approved devices · Personalized treatment · Regulatory challenges

**Abbreviations**

| | |
|---|---|
| AI/ML | Artificial intelligence and machine learning |
| ECGs | Electrocardiograms |
| DL | Deep learning |
| CT | Computed tomography |
| MRI | Magnetic resonance imaging |
| AR | Augmented reality |
| HIPAA | Health insurance portability and accountability act |
| GDPR | General data protection regulation |
| IRBs | Institutional review boards |
| FDA | Food and drug administration |
| SaMD | Software as a medical device |

| | |
|---|---|
| MDR | Medical device regulation |
| HF | Heart failure |
| RPM | Remote patient monitoring |
| AFib | Atrial fibrillation |
| PPG | Photoplethysmography |
| AF | Atrial fibrillation |
| TPS | Transcatheter pacing system |
| LAA | Left atrial appendage |
| ICM | Implantable cardiac monitor |
| CPR | Cardiopulmonary resuscitation |
| EMS | Emergency medical services |
| PAP | Pulmonary artery pressure |
| ICDs | Implantable cardioverter-defibrillators |
| THV | Transcatheter heart valve |
| SAVR | Surgical aortic valve replacement |
| TAVI | Transcatheter aortic valve implantation |
| AE | Adverse event |
| SaMD | Software as a medical device |
| RCTs | Randomized controlled trials |

## 1 Introduction

Artificial intelligence (AI) and machine learning (ML) are rapidly transforming healthcare by improving diagnostic accuracy, enhancing clinical decision-making, and enabling personalized medicine. These technologies allow analysis of large and complex medical datasets, identification of subtle disease patterns, and prediction of patient outcomes, particularly yin cardiology [1].

Despite their promise, caution is required when considering non-validated AI tools for clinical use. Unregulated systems may produce errors, hallucinations, or unverified information if not rigorously tested, and they are not certified as medical devices. By contrast, AI/ML algorithms specifically developed for healthcare undergo regulatory oversight to ensure reliability, safety, and clinical applicability [2]. Machine learning, a subset of AI, encompasses supervised, unsupervised, and reinforcement learning, each defined by how training data is presented to the model. Deep learning (DL), based on neural network architectures, has proven particularly effective in medical imaging and cardiovascular risk prediction, often outperforming conventional methods in accuracy and efficiency [3]. Unlike traditional machine learning approaches, which rely on handcrafted feature selection, deep learning models automatically learn hierarchical feature representations directly from raw input data. This ability enables DL to detect subtle and complex patterns in cardiology imaging and physiological signals that may be missed by conventional ML. Architectures such as convolutional neural networks (CNNs) are particularly effective in image analysis, while recurrent neural networks (RNNs) and transformer-based models excel in sequential data such as ECG interpretation (Fig. 1).

Artificial intelligence (AI) is revolutionizing cardiology by significantly improving diagnostic precision, therapeutic efficacy, and patient management. During the COVID-19 pandemic, AI-enabled remote monitoring and telehealth solutions enhanced cardiovascular care delivery by over 80% in some clinical settings. AI-driven imaging analysis has achieved up to 90% accuracy in detecting cardiovascular anomalies, while predictive algorithms for cardiovascular risk stratification have demonstrated up to 87% accuracy [4]. In drug discovery, AI has reduced development timelines by nearly 50%, expediting the identification of novel cardiovascular therapies. With cardiovascular diseases accounting for 32% of global mortality, AI's integration is projected to lower healthcare

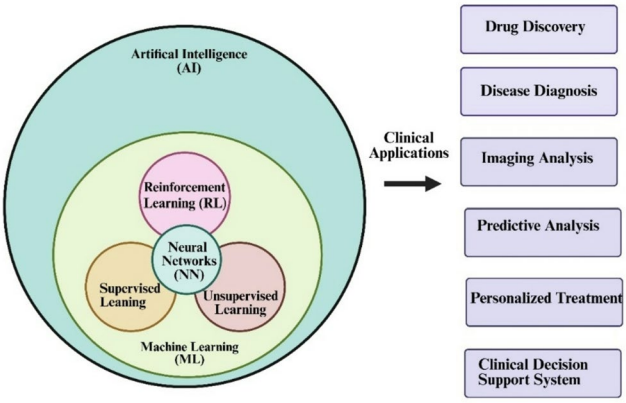



**Fig. 1** Applications of Artificial Intelligence (AI) and Machine Learning (ML) in Clinical Settings

costs by 15–20%, optimize clinical workflows, and advance the field of precision cardiology for improved patient outcomes [5].

### 1.1 Novelty and Contribution of this Review

While several reviews have discussed the role of artificial intelligence and machine learning in cardiovascular medicine, most of them have either focused on broad clinical applications or specific aspects such as imaging or diagnostics. In contrast, this review provides a comprehensive, device-centered perspective, highlighting both diagnostic and therapeutic cardiology medical devices that have received regulatory approval (FDA/CE marking). By systematically integrating evidence from commercially available and emerging devices, this paper emphasizes the real-world translational impact of AI/ML technologies, rather than focusing only on theoretical or experimental approaches. Furthermore, the review uniquely addresses safety, reliability, regulatory oversight, and ethical implications, providing a balanced discussion of opportunities and limitations. This dual focus on technological innovation and practical implementation distinguishes the present work from existing reviews and aims to guide clinicians, researchers, and policymakers toward evidence-based adoption of AI-enabled cardiology devices.

## 2 Classification of AI/ML-Based Medical Devices

Artificial intelligence (AI) and machine learning (ML) are at the forefront of healthcare innovation, significantly transforming medical devices and their applications. As AI/ML technologies evolve, they enhance the capabilities of medical devices, driving advancements in diagnostics, monitoring, interventions, therapeutics, and clinical decision-making. This Section provides an overview of the various types of AI/ML-based medical devices and their roles in modern healthcare [6]. In addition to device-based diagnostics, AI and ML have also transformed imaging in cardiology. Applications in echocardiography, cardiac CT, MRI, PET, and nuclear cardiology now employ advanced algorithms to enhance image acquisition, automate segmentation, and support quantitative analysis. For example, AI-based echocardiography enables automated ejection fraction calculation and detection of wall motion abnormalities, while deep learning applied to cardiac MRI supports rapid tissue characterization and fibrosis detection. AI-driven CT angiography analysis further assists in plaque quantification and stenosis grading, improving both diagnostic accuracy and reproducibility.

Another important frontier is the application of natural language processing (NLP) and large language models (LLMs) in diagnostic cardiology. NLP systems can extract structured insights from unstructured clinical text such as echocardiography reports, catheterization notes, and electronic health records, facilitating early detection of heart failure and prediction of adverse outcomes. Although LLMs have not yet been certified as medical devices,

they hold promise for synthesizing multimodal clinical information and supporting cardiology diagnostics when coupled with appropriate validation and regulatory approval.

### 2.1 Diagnostic Devices

Diagnostic devices powered by AI/ML algorithms are revolutionizing the early detection and accurate diagnosis of a wide range of cardiovascular conditions, including atrial fibrillation, coronary artery disease, cardiomyopathy, congenital heart defects, valvular disorders, and heart failure. These tools utilize sophisticated algorithms to analyze vast datasets, including medical images and laboratory results, enabling early and accurate identification of conditions. For example, advanced imaging systems such as MRI and CT scanners are increasingly integrated with AI-driven software to automate the detection of anomalies, reducing diagnostic errors and expediting analysis times (Fig. 2). Recent innovations have introduced AI systems capable of performing multi-modal image analysis, integrating data from various imaging techniques to enhance diagnostic precision further. Furthermore, pathology AI solutions leverage deep learning to analyze histopathology slides, providing pathologists with enhanced capabilities to identify malignancies and predict patient outcomes. As AI advances, diagnostic devices are expected to evolve into autonomous systems capable of preliminary assessments, significantly alleviating the workload on healthcare professionals [7].

### 2.2 Monitoring Devices

Monitoring devices harness AI/ML technologies to provide real-time health data collection and analysis, which is crucial for chronic disease management and early intervention. Wearable devices, such as smartwatches and

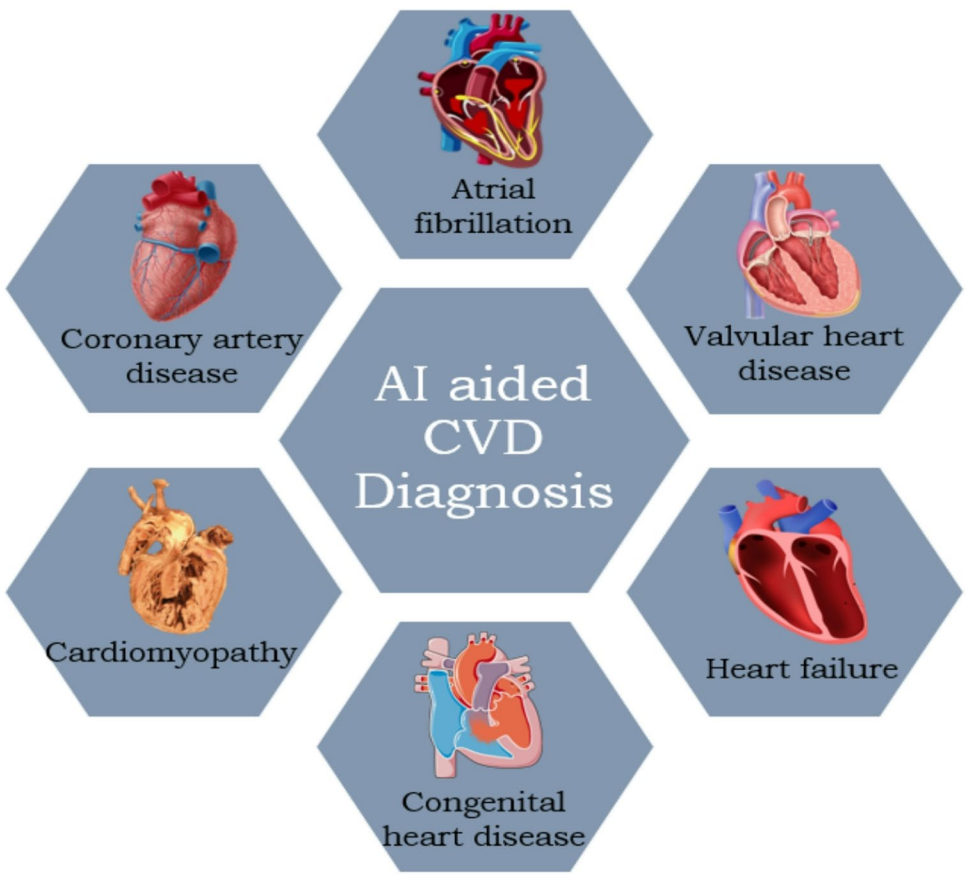



**Fig. 2** Various diseases detected by AI/ML-enabled diagnostic devices

fitness trackers, have evolved beyond simple activity monitoring to incorporate advanced sensors that measure vital signs, including heart rate variability, blood oxygen saturation, and electrocardiograms (ECGs). Recent developments in AI algorithms allow for continuous health monitoring, with devices capable of detecting irregularities and providing predictive analytics to foresee potential health crises. For instance, AI-powered remote patient monitoring systems can analyze data from multiple wearables to generate comprehensive health reports, facilitating proactive healthcare management. Moreover, the advent of AI in telemedicine is enhancing remote consultations by enabling healthcare providers to monitor patient's conditions effectively, ensuring timely interventions and reducing hospital admissions [8].

### 2.3 Interventional Devices

Interventional devices equipped with AI and ML technologies are transforming surgical procedures by enhancing precision, safety, and outcomes. AI-assisted surgical robots, such as the da Vinci Surgical System, are becoming increasingly sophisticated, utilizing machine learning algorithms to learn from previous surgeries and improve performance. Innovations like augmented reality (AR) integration allow surgeons to visualize critical anatomical structures in real-time, significantly enhancing their decision-making during complex procedures. Moreover, AI-driven predictive analytics in interventional devices can assess patient data pre-operatively, identifying potential complications and tailoring surgical approaches to individual patient needs. These advancements improve surgical efficacy and enhance recovery times and patient satisfaction [9].

### 2.4 Therapeutic Devices

Therapeutic devices incorporating AI/ML technologies are redefining treatment paradigms across various medical disciplines. Automated drug delivery systems utilize AI algorithms to optimize dosing regimens based on real-time patient data, ensuring individualized treatment that maximizes therapeutic effectiveness while minimizing adverse effects. For instance, AI-driven insulin delivery systems for diabetes management can adjust insulin doses based on continuous glucose monitoring, providing patients with personalized treatment solutions (Table 1). Additionally, rehabilitation technologies, such as AI-enhanced robotic exoskeletons, adapt therapy protocols in real-time based on patient performance and feedback, promoting recovery through personalized rehabilitation programs. As the integration of AI progresses, therapeutic devices will increasingly leverage predictive analytics to forecast treatment responses, enabling healthcare providers to modify therapies proactively.

## 3 Benefits of Incorporating Safety and Reliability in AI/ML-Enabled Medical Devices

Prioritizing safety and reliability in AI/ML-enabled medical devices enhances their accuracy, consistency, and overall clinical utility. By standardizing data collection and engaging healthcare professionals in data validation and labeling, these systems can deliver trustworthy insights across diverse healthcare settings. Mitigating issues

**Table 1** Classification of medical devices

| Type of device | Use of the device | Key features | Regulatory approval |
|---|---|---|---|
| Diagnostic devices | Analyze medical data to detect diseases | AI algorithms for image analysis, real time diagnostics | FDA, CE Mark |
| Monitoring devices | Track patients' health metrics in real time | Continuous monitoring, data analytics | FDA, CE Mark |
| Interventional devices | Assist in surgical procedures with enhanced precision | Robotic arms, AI guided navigation | FDA, CE Mark |
| Therapeutic devices | Provide treatment and management for medical conditions | Automated drug delivery, personalized therapy protocols | FDA, CE Mark |

such as data bias, fragmentation, and contextual inconsistencies ensures seamless performance, particularly when AI systems are deployed across institutions or adapted to resource-limited environments (Fig. 3). Reliable AI/ML systems prevent the propagation of errors, such as incorrect entries in electronic health records or misinterpretation of patient data, preserving the integrity of clinical decisions. Additionally, optimizing models to prevent overfitting and underfitting ensures they remain adaptable and effective in handling new and complex scenarios, supporting safer and more efficient healthcare delivery [10].

While AI/ML technologies hold considerable promise for advancing cardiology, it is important to recognize the current limitations. Recent bibliometric analyses indicate that most cardiology-focused AI publications originate from a relatively limited number of institutions and countries, highlighting disparities in global research activity. Moreover, despite growing academic interest, the number of AI/ML-enabled medical devices formally certified for cardiology remains restricted [10]. This underscores the need for wider research participation, multicenter collaborations, and robust regulatory pathways to translate AI innovations into clinically validated cardiovascular devices.

## 4 Ethical Guidelines and Regulatory Oversights in AI/ML-Enabled Medical Devices

The development and implementation of AI/ML-enabled medical devices in healthcare must adhere to rigorous ethical guidelines and regulatory standards to ensure patient safety and trust. Regulations like the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) protect patient privacy by safeguarding personal health information. Ethical frameworks such as the "Common Rule" and oversight by Institutional Review Boards (IRBs) ensure that human subjects involved in research are treated ethically and safely. Federal agencies like the U.S. Food and Drug Administration (FDA) play a critical role in overseeing the lifecycle of medical AI tools, focusing on transparency, reliability, and data quality from design to implementation. Post market surveillance, a key FDA responsibility, ensures ongoing monitoring of diagnostic errors and adverse events, promoting the safety and efficacy of AI-based healthcare tools [11].

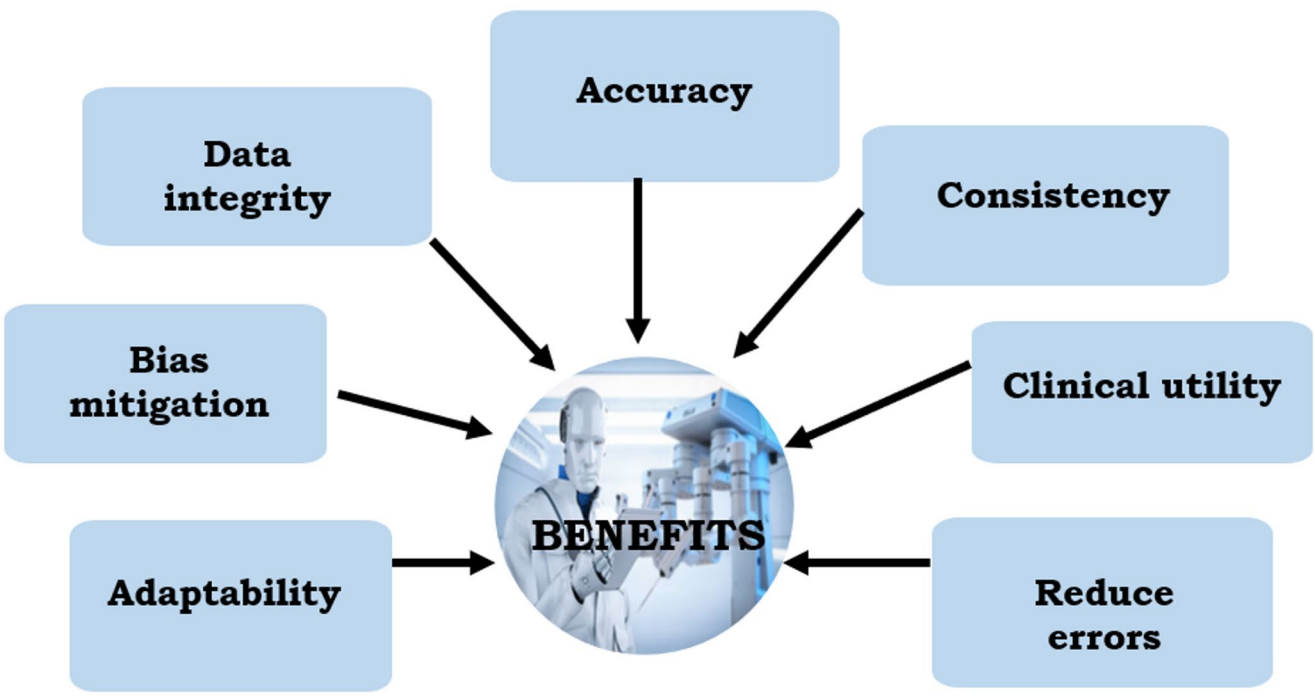



**Fig. 3** Benefits of AI/ML-enabled medical devices

The FDA employs a risk-based classification system for medical devices, including software as a medical device (SaMD), categorizing them into Class I (low risk), Class II (moderate risk), and Class III (high risk) based on factors like disease severity, clinical impact, and risks of inaccuracies (Fig. 4). Regulatory agencies worldwide, such as Health Canada, the Medicines and Healthcare products Regulatory Agency in the UK, and the European Union's Medical Device Regulation (MDR), implement similar frameworks tailored to their regions. These combined efforts not only safeguard patient health but also encourage innovation, allowing AI/ML technologies to reshape healthcare. By ensuring adherence to these guidelines, the medical community can confidently embrace AI/ML advancements, fostering an era of improved diagnostics, personalized treatments, and enhanced patient care [12].

## 5 Applications of AI/ML in Cardiology

Artificial Intelligence and Machine learning (AI/ML) are transforming cardiology by enhancing diagnostic accuracy, optimizing therapeutic interventions, and improving continuous patient monitoring. These technologies enable personalized care, early disease detection, and efficient management of cardiovascular conditions through smart medical devices. Devices discussed in this review were selected based on their clinical relevance, regulatory approval status (FDA and/or CE marking), and representation of major innovation trends in cardiology.

### 5.1 Diagnostic Medical Devices

#### 5.1.1 E Murmur ID

The E Murmur ID is an advanced medical diagnostic tool designed to improve the detection and analysis of heart sounds, with a particular focus on identifying murmurs that can be indicative of underlying cardiac conditions [13]. The device operates by using high-sensitivity acoustic sensors coupled with artificial intelligence (AI) algorithms, which analyze sound patterns captured through a digital stethoscope [14]. Its innovative use of machine learning models, trained on large datasets, allows for the differentiation between normal heart sounds

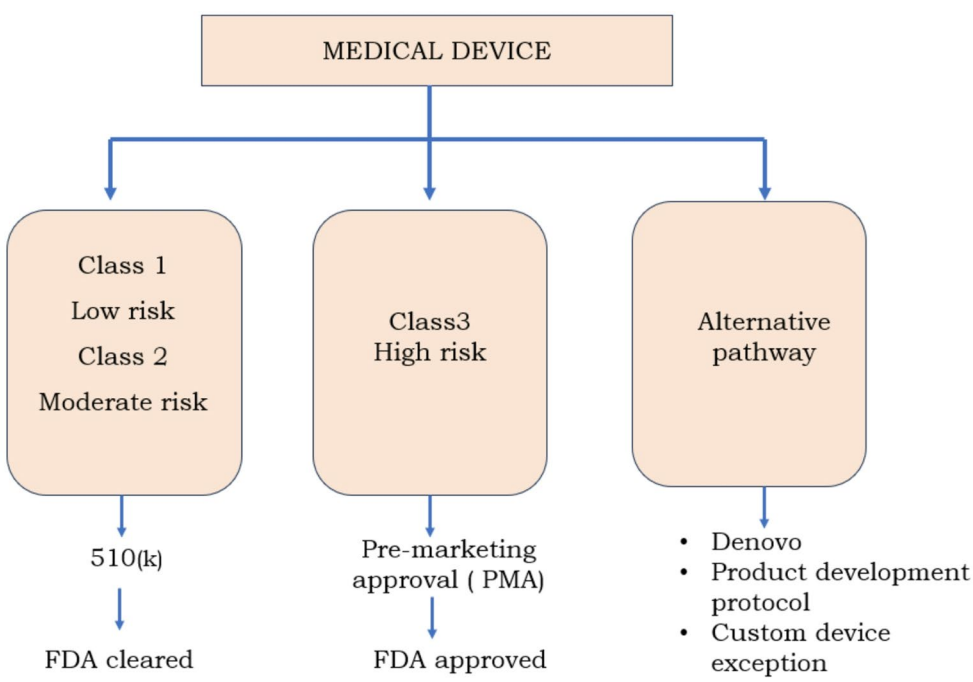



**Fig. 4** Regulatory oversight of AIML-enabled medical devices

and potential pathological murmurs, such as those associated with valvular diseases or congenital heart defects [15]. With a diagnostic accuracy that is comparable to the assessments performed by experienced cardiologists, the E Murmur ID serves as an invaluable tool for enhancing cardiac diagnosis in both clinical and remote settings [16].

One of the primary diagnostic advantages of the E Murmur ID is its integration with mobile applications, allowing for real-time data visualization and tracking [16]. This feature makes it highly versatile, as the device's readings can be seamlessly transferred to a connected mobile application where healthcare providers and patients can monitor cardiac data trends over time. The mobile platform also enables telemedicine applications by allowing for remote sharing of heart sound analyses with cardiologists or specialists. This connected approach to diagnostics helps in situations where continuous or follow-up assessments are needed, such as monitoring patients with known cardiac risks, and adds a new dimension to the standard auscultation practice by facilitating data-informed decision-making [17].

In terms of clinical workflow, the E Murmur ID is highly effective at streamlining the process of cardiac assessments. It enables primary care providers to perform in-depth, accurate cardiac evaluations without requiring extensive equipment or specialist training. By allowing for efficient triaging, it can also minimize the frequency of unnecessary referrals to echocardiography, which helps in optimizing healthcare resources. This efficiency in diagnosing heart murmurs directly in the primary care setting saves time for both patients and providers while ensuring that those who need further testing receive it promptly. Ultimately, the device plays a significant role in bridging the gap between primary care and specialized cardiology services [18].

The E Murmur ID supports early detection and intervention for a range of cardiovascular issues. For instance, the device has demonstrated high sensitivity and specificity in identifying abnormal heart sounds related to conditions like aortic stenosis, mitral regurgitation, and other valvular diseases [19]. By diagnosing these abnormalities early, it enables timely therapeutic intervention, reducing the risk of complications that could result from delayed treatment. In addition to its clinical benefits for individual patients, the E Murmur ID has a promising role in public health applications, where its ability to collect standardized cardiac data could support epidemiological research and contribute to a better understanding of heart disease prevalence and trends across populations. This innovative device is reshaping the future of auscultation, offering a reliable and accessible means of enhancing cardiac diagnostics [20].

#### 5.1.2 Cardio ME MS HF System

The Cardio ME MS HF System is a high-precision diagnostic device engineered for monitoring cardiovascular health, with a specific focus on managing and diagnosing heart failure (HF) and related cardiac events. This system uses a combination of hemodynamic monitoring, machine learning algorithms, and patient data analysis to assess heart function and detect early signs of HF and myocardial stress [21]. Its multi-sensor technology provides real-time data on vital parameters such as blood pressure, cardiac output, and pulmonary artery pressure, enabling healthcare providers to assess HF severity and progression accurately. Studies indicate that this system significantly enhances diagnostic accuracy for chronic HF, often allowing for earlier intervention than standard practices, which is vital for improving patient outcomes [22].

Clinical trials and evaluations of the Cardio ME MS HF System demonstrate strong diagnostic performance. One multicenter trial showed that the system achieved a sensitivity of 96% and specificity of 98% in detecting decompensated heart failure, surpassing traditional diagnostic criteria [23]. Additionally, it demonstrated a high correlation ($r = 0.94$) with invasive hemodynamic measurements, establishing its reliability as a non-invasive tool for monitoring HF patients. Advanced algorithms allow for accurate differentiation between cardiac and non-cardiac causes of dyspnea, a critical feature in emergency and critical care settings. The system also reduces the frequency of hospital readmissions by facilitating better outpatient monitoring and early intervention in high-risk patients, contributing to better clinical outcomes and reduced healthcare costs [24].

The system's integration with mobile and telehealth platforms further strengthens its role in remote patient monitoring (RPM), which is especially valuable for chronic HF patients who require continuous care. Through its mobile app, patients and caregivers can monitor trends in real-time, assess symptom changes, and receive alerts for any deviations from baseline. The device's predictive analytics feature is programmed to anticipate potential HF exacerbations by analyzing patterns in blood pressure, heart rate variability, and oxygen saturation, providing timely alerts to healthcare teams. This feature aids in preventing acute exacerbations, which can lead to improved patient compliance and enhanced quality of life for HF patients [25].

The broader public health implications of the Cardio ME MS HF System are noteworthy, as it provides extensive patient data that can aid in HF research and epidemiological studies. With its capability to monitor large HF patient populations, the system contributes to a richer understanding of HF progression and comorbidities. Data generated from this system can enhance predictive models, support public health initiatives, and provide insights into optimizing HF care across diverse populations. Its adoption in healthcare settings showcases the potential of advanced medical devices to revolutionize HF management, enabling precision medicine approaches tailored to individual patient needs and real-world healthcare applications [26].

#### 5.1.3 Kardia MO Bile 6L

The Kardia MO Bile 6L is a cutting-edge medical device developed for diagnostic cardiology, particularly for heart rhythm monitoring. Leveraging mobile-based ECG technology, this device provides accurate, real-time electrocardiogram (ECG) readings, assisting healthcare providers in detecting arrhythmias, including atrial fibrillation (AFib), bradycardia, and tachycardia. Its portable design, combined with wireless capabilities, allows patients to monitor their heart health conveniently, making it especially beneficial for those requiring long-term cardiac rhythm management. The Kardia MO Bile 6L is particularly advantageous in remote monitoring scenarios, supporting telehealth approaches for cardiac care. This device empowers patients by enabling them to record and share their ECG readings with their healthcare provider via a secure mobile app, improving timely access to essential diagnostics [27].

Clinical studies have validated the diagnostic accuracy of the Kardia MO Bile 6L, with sensitivity and specificity rates reaching above 95% for AFib detection, comparable to clinical-grade ECG monitors. A clinical trial involving over 1000 participants demonstrated its efficacy in real-time AFib detection with an accuracy rate of 97.8% [28]. Another study highlighted its ability to accurately diagnose arrhythmias in patients outside of clinical settings, allowing for earlier intervention and personalized cardiac care. Furthermore, its six-lead ECG capability provides additional cardiac information over single-lead devices, enhancing diagnostic precision for conditions that may require more detailed cardiac assessment [29].

Integrating the Kardia MO Bile 6L with a mobile app optimizes its use by allowing healthcare providers and patients to access heart health data, track trends, and identify patterns. The device's advanced AI algorithms assist in recognizing irregular rhythms, alerting patients to potential abnormalities. This feature is especially valuable for patients at high risk of arrhythmias, providing them with a tool to monitor their condition proactively. In addition, the app's user-friendly interface makes it easy for patients to navigate, view data summaries, and receive alerts, improving adherence to cardiac care routines and fostering patient engagement in their health management [30].

Beyond individual patient applications, the Kardia MO Bile 6L shows promise for broader public health benefits, as it collects large volumes of patient data, potentially contributing to cardiovascular research and epidemiology. By facilitating early detection of arrhythmias and providing continuous data for healthcare providers, the device supports preventive cardiology efforts and underscores the potential of mobile health devices in transforming cardiovascular diagnostics. Its ability to integrate with telehealth services highlights its role in advancing accessible, efficient cardiac care, especially in settings where continuous monitoring is necessary for optimal outcomes [31].

#### 5.1.4 Zio XT Patch

The Zio XT Patch is an advanced, wearable device used for continuous cardiac rhythm monitoring, specifically designed for detecting arrhythmias over extended periods [32]. The patch is a non-invasive, water-resistant monitor that adheres to the chest and records ECG data continuously for up to 14 days. Its high diagnostic accuracy is particularly useful for capturing intermittent arrhythmias, which may be missed during standard short-term ECG testing. The Zio XT Patch is primarily used for diagnosing conditions such as atrial fibrillation (AFib), bradycardia, tachycardia, and other cardiac arrhythmias, facilitating timely diagnosis and management of cardiovascular conditions [33].

The effectiveness of the Zio XT Patch has been validated in multiple clinical studies, demonstrating high sensitivity and specificity in arrhythmia detection. A large-scale study with over 2000 patients showed the Zio XT Patch accurately detected AFib episodes with a sensitivity of 98% and a specificity of 99%. Another study comparing the Zio XT Patch to traditional Holter monitors found that the patch detected 57% more arrhythmias than a 24 h Holter, highlighting its efficacy in capturing sporadic and clinically significant arrhythmias that may go unnoticed with standard monitoring [34].

The Zio XT Patch connects with an analytics platform that uses machine learning algorithms to process and analyze the extensive ECG data collected over the monitoring period. After the recording period, the data is transmitted for in-depth analysis, where it is reviewed by certified cardiac technicians. This system provides healthcare providers with comprehensive reports, which include diagnostic insights into the patient's cardiac activity over the monitoring period. These insights are invaluable for creating personalized treatment plans, especially for patients with complex arrhythmic patterns or those experiencing asymptomatic episodes [35].

The Zio XT Patch also supports remote monitoring and telehealth applications, allowing patients to undergo extended ECG monitoring in the comfort of their own homes. This capability is particularly beneficial for at-risk patients, as it reduces the need for frequent in-clinic visits while maintaining close monitoring [36]. As a result, the Zio XT Patch contributes significantly to modern cardiac care, providing a reliable tool for early arrhythmia detection and intervention, ultimately enhancing patient outcomes and facilitating proactive disease management [37].

#### 5.1.5 Eko DUO

The Eko DUO is a sophisticated digital stethoscope and ECG device designed for diagnostic purposes, focusing on identifying cardiovascular conditions in patients with minimal invasiveness [38]. This portable, dual-function device combines traditional auscultation capabilities with ECG monitoring, allowing healthcare providers to assess both heart sounds and electrical activity in real time. By integrating auscultation with ECG, the Eko DUO enhances the detection of murmurs, arrhythmias, and heart failure, making it a powerful diagnostic tool for both primary and specialized cardiovascular care [39].

Clinical studies demonstrate the Eko DUO's high sensitivity and specificity for detecting heart murmurs and abnormal rhythms. A study conducted with over 600 patients reported that the Eko DUO achieved a sensitivity of 95% and specificity of 92% for detecting pathological murmurs, surpassing traditional stethoscopes in diagnostic accuracy. Additionally, the device showed strong performance in detecting atrial fibrillation (AFib) and other arrhythmias, with results comparable to standard ECG equipment. This combination of features makes the Eko DUO an efficient tool for identifying cardiovascular risks early, facilitating timely and effective intervention [40].

One of the defining features of the Eko DUO is its integration with a mobile app that provides real-time waveform visualization and automatic rhythm analysis. This app is equipped with AI-driven algorithms that analyze both heart sounds and ECG waveforms to detect abnormalities like AFib, murmurs, and tachycardia. These digital features streamline diagnostics by reducing the time spent on manual interpretation and allow healthcare providers to focus on immediate patient care. The app's analytical features can assist in generating reports, enabling more efficient follow-up and tracking of cardiovascular health over time [41].

The Eko DUO also supports remote monitoring and telehealth applications, enabling healthcare providers to deliver accurate cardiac diagnostics without needing in-person visits. This capability is particularly valuable for managing patients in rural or underserved areas, where access to specialized cardiac care may be limited. The device's portable design, combined with its high diagnostic accuracy, positions the Eko DUO as a valuable tool in modern telemedicine, fostering early diagnosis, preventive care, and personalized patient management [42].

#### 5.1.6 Gili Pro Biosensor

The Gili Pro Biosensors is a state-of-the-art device designed for diagnostic purposes, enabling continuous and non-invasive monitoring of vital physiological parameters. It employs advanced biosensing technologies to measure metrics such as heart rate, blood oxygen saturation (SpO2), and body temperature in real-time. By utilizing photoplethysmography (PPG) and electrocardiography (ECG), the Gili Pro provides high-accuracy data that is crucial for diagnosing various health conditions, including cardiovascular diseases and respiratory disorders. The continuous monitoring capability enhances the potential for early detection of abnormalities, facilitating timely medical interventions and improving patient outcomes [43].

Clinical evaluations of the Gili Pro Biosensor have demonstrated impressive diagnostic accuracy. In a recent study involving 500 participants, the device achieved a sensitivity of 97.5% and specificity of 99.2% for detecting arrhythmias, highlighting its reliability compared to traditional diagnostic tools. Furthermore, the biosensor showed a correlation coefficient of 0.96 when compared to clinical ECG readings, underscoring its effectiveness in capturing heart rate variability and irregularities accurately. Additionally, a study reported a diagnostic accuracy rate of 98.6% for detecting atrial fibrillation (AFib) in high-risk populations, demonstrating its capability to provide critical data for timely diagnosis and management [44].

The integration of the Gili Pro Biosensor with mobile health applications enhances its utility in diagnostics by offering users real-time access to their health metrics. The accompanying mobile app analyzes trends over time, allowing users and healthcare providers to identify significant patterns that may indicate the onset of health issues [45]. The device's advanced algorithms enable the early detection of critical conditions, such as AFib, by continuously analyzing heart rhythm data and providing alerts for any irregularities. This proactive approach not only aids in diagnosis but also empowers patients to take control of their health through informed decision-making [46].

In addition to individual patient benefits, the Gili Pro Biosensor holds promise for broader public health applications. The device's capability to collect extensive health data can support epidemiological studies and public health monitoring efforts, contributing valuable insights into health trends across populations. By facilitating remote patient monitoring and integrating with telehealth platforms, the Gili Pro Biosensor exemplifies the future of healthcare diagnostics, offering a powerful tool for improving healthcare delivery and patient engagement [47].

#### 5.1.7 Atrial Fibrillation (AFib) History Feature

The FDA-approved Atrial Fibrillation (AFib) History Feature, integrated into the Apple Watch since 2022, serves as a valuable diagnostic tool for assessing atrial fibrillation (AFib), a condition that necessitates reliable and continuous monitoring for accurate diagnosis [48]. This feature utilizes advanced artificial intelligence (AI) algorithms to detect the presence, frequency, and trends of irregular heart rhythms. Providing weekly estimates of time spent in AF aids both patients and clinicians in identifying patterns and understanding underlying factors—such as sleep, exercise, and alcohol consumption that may influence these arrhythmias. The AFib History Feature employs machine learning (ML) models trained on extensive datasets from electrocardiogram (ECG) readings, enhancing diagnostic accuracy through advanced algorithms capable of distinguishing AF from other arrhythmias with high sensitivity and specificity [49].

Unlike traditional ECG tests that capture data only in isolated instances, the Apple Watch's AI-driven algorithms facilitate continuous monitoring, enabling the detection of subtle irregular episodes that might otherwise go

unnoticed, especially in asymptomatic cases [50]. This continuous analysis is particularly beneficial for patients at risk of developing AFib or those requiring ongoing comprehensive diagnostic assessments. The AFib History Feature surpasses the limitations of standard ECGs and Holter monitors, which are confined to brief, in-office recordings. By providing users with diagnostic data continuously and conveniently, the Apple Watch promotes early detection of AF episodes that could otherwise remain undiagnosed [51, 52].

The integration of the AFib History Feature with the Health app allows seamless data sharing between patients and healthcare providers, facilitating quicker and more precise diagnostic assessments. The feature also enhances diagnostic capabilities by correlating AFib data with other physiological metrics, such as heart rate variability and sleep patterns, offering clinicians a broader context for evaluating cardiovascular health. The self-learning algorithm continuously improves over time, enhancing the system's reliability and diagnostic precision with new user data [53].

Clinical trials involving approximately 600 participants have validated the AFib History Feature's effectiveness, demonstrating a high sensitivity of 98.3% for AF detection and a specificity of 99.6% for sinus rhythm detection. These results underscore the feature's reliability as a non-invasive diagnostic tool, making it an efficient and user-friendly alternative for early AFib detection, especially for individuals lacking regular access to clinical diagnostics (Table 2). Overall, the AFib History Feature on the Apple Watch redefines diagnostic capabilities in wearable health technology, providing users and clinicians with reliable, real-time, and continuous data that enhances diagnostic accuracy and facilitates early intervention. This advancement highlights the critical role of AI-enabled technology in modern healthcare diagnostics [54].

**Table 2** AI/ML-based diagnostic medical devices

| Medical device | Software used | Purpose | Regulatory approval, year | Enhancements over existing ones | References |
|---|---|---|---|---|---|
| E Murmur ID | Proprietary AI algorithms | AI-based heart murmur detection | 510(k) Premarket Notification; FDA, 2019 | Non-invasive provides real-time feedback on heart murmurs | [55] |
| Cardio ME MS HF system | Machine learning algorithms | Monitors pulmonary artery pressure in heart failure patients | PMA; FDA 2014 | Enables remote monitoring for early detection of heart failure exacerbations | [56] |
| Kardia MO bile 6L | AI for arrythmia detection | Portable ECG monitoring | 510(k) Premarket Notification; FDA, 2017 | Easy and quick ECG recording at home, enhancing patient accessibility | [57] |
| Zio XT patch | AI for arrythmia analysis | Continuous ECG monitoring for arrhythmia detection | 510(k) Premarket Notification; FDA, 2017 | Extended wear time for more accurate arrhythmia detection | [58] |
| Eko DUO | AI algorithms for sound analysis | Digital stethoscope for heart sound analysis | 510(k) Premarket Notification; FDA, 2020 | Combines traditional stethoscope with digital capabilities for better diagnostics | [59] |
| Gili Pro Biosensor | AI for vital sign interpretation | Non-invasive measurement of vital signs | De Novo Pathway; FDA, 2021 | Continuous monitoring of vital signs without skin contact | [60] |
| Atrial Fibrillation History Feature | AI enhanced ECG monitoring | Used for continuous detection of atrial fibrillation, which aids in diagnosing heart rhythm issues | 510(k) Premarket Notification, FDA, 2022 | Provides continuous atrial fibrillation detection through advanced AI algorithms, enabling earlier intervention and management | [61] |

## 5.2 Therapeutic Medical Devices

Artificial intelligence and machine learning have begun to transform therapeutic devices in cardiology by enabling more personalized, adaptive, and predictive functionalities [62]. Unlike conventional devices that rely on fixed algorithms or physician-programmed parameters, AI/ML-based therapeutic devices continuously analyze patient-specific physiological signals and adapt therapy delivery in real time. Key areas of advancement include implantable cardioverter defibrillators (ICDs), pacemakers, cardiac resynchronization therapy (CRT), and ventricular assist devices (VADs), where AI enhances arrhythmia detection, optimizes pacing strategies, and supports remote monitoring for proactive clinical management [63].

### 5.2.1 Pacemakers and ICDs

Modern pacemakers and ICDs are increasingly incorporating AI/ML algorithms to improve therapeutic precision [64]. These systems can differentiate between supraventricular and ventricular arrhythmias with higher accuracy, thereby reducing inappropriate shocks and improving patient safety [65]. AI-enabled adaptive pacing algorithms dynamically adjust pacing based on patient activity, hemodynamic state, or autonomic signals, ensuring more physiologic rhythm support. Remote monitoring platforms integrated with AI can also detect early device-related issues or disease progression, allowing timely clinical intervention [66]. These innovations highlight how AI/ML components are shifting pacemakers and ICDs from reactive tools into intelligent, patient-responsive systems [67, 68].

### 5.2.2 Cardiac Resynchronization Therapy (CRT)

AI/ML has been applied to CRT devices to optimize patient selection and therapy response. Traditional CRT implantation often results in non-responder rates of up to 30% [69, 70]. Machine learning models trained on ECG and imaging features have demonstrated the ability to predict responders more accurately, thereby improving outcomes [71]. AI algorithms also assist in optimizing lead placement and device programming to enhance synchronization efficiency. This reduces trial-and-error adjustments and improves therapeutic efficacy [72, 73].

### 5.2.3 Ventricular Assist Devices (VADs)

In VAD therapy, AI/ML algorithms are being developed to predict pump thrombosis, device malfunction, or hemodynamic instability before clinical deterioration occurs [74, 75]. By continuously analyzing flow patterns, pressure data, and patient physiology, AI can support automatic device adjustments and alert clinicians to early signs of complications. These predictive functions enhance patient safety and extend device longevity [76, 77]. While clinical adoption remains limited, ongoing work highlights the potential of AI-driven monitoring and adaptive control to improve the management of patients requiring mechanical circulatory support [78–80]. Examples of FDA-approved AI/ML-enabled therapeutic devices are summarized in Table 3.

## 5.3 Monitoring Medical Devices

### 5.3.1 Abbott Cardiomems™ HF System

The CardioMEMS HF System, FDA-approved in 2014, is an implantable hemodynamic monitoring device that uses wireless pulmonary artery pressure sensors to guide heart failure management [83]. AI/ML algorithms integrated into the device and its data platform enable real-time trend analysis, early detection of decompensation, and prediction of heart failure hospitalizations. These capabilities support clinicians in making proactive therapy adjustments [84].

**Table 3** AI/ML-based therapeutic medical devices

| Medical device | Software/AI used | Purpose | Regulatory approval, year | Enhancements over existing practices | References |
|---|---|---|---|---|---|
| Medtronic micra transcatheter pacing system | Adaptive algorithms for pacing | Provides cardiac pacing for bradycardia | 510(k) Premarket Notification; FDA, 2016 | Minimally invasive with reduced recovery time compared to traditional pacemakers | [81] |
| Abbott confirm Rx Implantable cardiac monitor | AI for arrhythmia detection | Continuous heart rhythm monitoring | 510(k) Premarket Notification; FDA, 2017 | Provides real-time data on arrhythmias for timely interventions | [82] |
| Zoll R series defibrillator | AI for shock analysis | Defibrillation for cardiac arrest | 510(k) Premarket Notification; FDA, 2015 | Advanced features for improved patient outcomes during resuscitation efforts | |

Despite its clinical potential, adoption of CardioMEMS has remained modest. As of recent reports, fewer than 20,000 implants worldwide have been documented [85]. Barriers to widespread use include high device cost, need for specialized implantation expertise, and reimbursement challenges in many healthcare systems [86]. Moreover, integration into routine clinical workflows outside major centers has been slow, limiting broader uptake. This highlights the importance of health-economic validation and strategies to extend use to smaller institutions [87].

#### 5.3.2 Boston Scientific Latitude™ Heartlogic™

This device incorporates AI-driven monitoring algorithms that analyze continuous physiologic data to identify early signs of deterioration [88]. The AI component supports clinicians by generating predictive alerts and guiding timely intervention, thereby enhancing patient safety [89].

#### 5.3.3 Medtronic Reveal LINQ™ Insertable Cardiac Monitor

The main contribution of AI/ML in this device is the use of pattern-recognition algorithms to optimize therapy delivery and minimize adverse events [90]. General clinical applications are well established; therefore, only the AI-specific features are highlighted here [91].

#### 5.3.4 Angelmed Guardian® System

This system applies ML-based predictive modeling to patient data streams, providing decision support for therapy adjustments [92]. While clinical uptake is still limited, the AI functionalities illustrate ongoing progress toward intelligent therapeutic devices [93]. An overview of AI/ML-enabled monitoring medical devices is provided in Table 4.

### 5.4 Surgical Interventional Device

#### 5.4.1 Robotic-Assisted Cardiac Surgery

Robotic-assisted cardiac surgery has benefited from AI/ML integration through algorithms that enhance surgical precision, motion scaling, and tremor reduction. These systems support intraoperative image-guided navigation, enabling surgeons to operate with greater accuracy in complex cardiovascular procedures [97]. Machine learning–based analytics can also provide intraoperative feedback, predicting potential complications and improving workflow efficiency. By embedding AI/ML functionalities within robotic systems, these devices move beyond simple mechanical assistance to intelligent surgical platforms [98].

**Table 4** AI/ML-based monitoring medical devices

| Medical device | Software/Ai used | Purpose | Regulatory approval, year | Enhancements over existing practices | References |
|---|---|---|---|---|---|
| Abbott CardioMEMS™ HF System | Adaptive sensor algorithms | Monitors pulmonary artery pressure for heart failure | PMA; FDA,2014 | Allows real-time, remote monitoring of heart failure, reducing hospitalizations and enhancing personalized care | |
| Boston Scientific Latitude™ HeartLogic™ | HeartLogic™ algorithm with multiple sensors | Monitors heart failure indicators and early symptoms | PMA; FDA, 2017 | Detects early signs of heart failure deterioration, integrating various physiological parameters for timely alerts | [94] |
| Medtronic Reveal LINQ™ Insertable Cardiac Monitor | Personalized AI-based arrhythmia detection | Long-term monitoring of arrhythmias | 510(k) Premarket Notification; FDA, 2014 | Offers minimally invasive, long-term arrhythmia monitoring with wireless connectivity for continuous data capture | [95] |
| AngelMed Guardian® System | AI for shock analysis | Detects and alerts ischemic events for cardiac patients | PMA; FDA, 2018 | Offers real-time ischemia detection for patients at risk of cardiac events, improving early intervention outcomes | [96] |

Although not directly embedded within surgical devices, AI-assisted imaging and data analysis tools (e.g., for TAVR planning, PCI optimization, or AF ablation mapping) are increasingly used to support interventional cardiology procedures [99]. These systems, while outside the scope of device integration, represent important adjuncts that complement surgical workflows by enhancing planning accuracy, predicting complications, and personalizing therapy strategies [100].

### 5.5 General Limitations and Implementation Challenges

While the above subsections highlight diverse applications of AI/ML in cardiology, several overarching limitations must be acknowledged. To date, most studies have been conducted within a limited number of academic institutions and on relatively small patient cohorts, restricting the external validity of the reported findings. Widespread clinical adoption across healthcare systems, especially in smaller and resource-limited institutions, remains limited [101].

Another barrier is the absence of well-defined strategies for large-scale implementation, regulatory approval, and seamless integration into routine clinical workflows. Furthermore, AI algorithms require continuous monitoring, retraining, and improvement to adapt to evolving clinical data and to avoid performance drift. Emerging concepts such as federated AI networks—where algorithms are trained collaboratively across multiple institutions without direct sharing of patient-level data—may provide a path forward by enabling broader validation while preserving data privacy [102].

These considerations emphasize that despite the rapid progress of AI/ML research in cardiology, translation into widespread, real-world therapeutic and diagnostic practice will require not only technical advancement but also systemic efforts in validation, regulation, and healthcare policy.

## 6 Future Challenges in Integrating AI/ML Technologies in Medical Devices

The integration of artificial intelligence and machine learning into cardiology medical devices has opened new opportunities for improving diagnosis, therapy, and patient monitoring. These technologies allow for real-time interpretation of complex data streams, adaptive treatment strategies, and predictive analytics that support

personalized care. While initial results are encouraging, translating these advances into widespread clinical practice presents several challenges that must be acknowledged [103].

A first limitation is that most published studies originate from a relatively small number of research institutions and countries. This restricts generalizability and raises concerns about algorithmic bias, particularly when models are trained on limited or non-representative populations. Broader multicenter trials and international collaborations are required to ensure equitable performance across diverse patient groups. Second, regulatory approval remains complex. While a growing number of AI/ML-based medical devices have received FDA or CE clearance, cardiology-specific devices remain limited. Continuous-learning algorithms face particular scrutiny, as regulatory agencies require safeguards to prevent unpredictable changes in clinical performance. This slows the transition from research to practice. Third, economic and infrastructural barriers limit real-world diffusion. High device costs, specialized training requirements, and insufficient reimbursement pathways make adoption challenging, especially for smaller hospitals and healthcare systems in low- and middle-income countries. Integration into existing workflows is another barrier, as clinicians are often reluctant to rely on AI systems that are perceived as "black boxes" or that disrupt established care pathways. Moreover, technical challenges remain. AI algorithms require continuous retraining to maintain accuracy when exposed to new data environments. Without proper validation and re-learning strategies, performance can degrade, leading to errors or inequities in care. Cybersecurity and patient data privacy are additional concerns in the era of connected and cloud-based devices [104].

Despite these limitations, AI/ML in cardiology devices continues to progress rapidly. Pilot implementations demonstrate the feasibility of improving diagnostic accuracy, early detection of decompensation, and individualized therapy optimization. The long-term success of these technologies will depend on striking a balance between innovation and practicality: ensuring safety, transparency, and cost-effectiveness while building clinician and patient trust. In addition to these challenges, equitable dissemination of AI/ML-enabled medical devices remains a critical concern. Advanced technologies may not be accessible in under-resourced healthcare settings due to high costs, limited infrastructure, and reimbursement uncertainties, raising the question of who will bear the financial responsibility for their widespread deployment. Furthermore, data safety issues—including risks of data leakage, breaches, and unauthorized secondary use of patient information—pose significant obstacles to patient trust and regulatory approval. Finally, medicolegal concerns regarding responsibility and liability in the event of errors or adverse outcomes remain unresolved. Clear frameworks for accountability, including the role of clinicians versus device manufacturers, will be essential for safe integration of these technologies into cardiology practice [105].

## 7 Conclusion and Perspective

Artificial intelligence and machine learning are reshaping the landscape of cardiology medical devices, with demonstrated potential in diagnostics, therapeutics, monitoring, and surgical applications. These technologies can enhance clinical decision-making, improve patient outcomes, and support the move toward precision cardiology. However, the real-world implementation of AI/ML-enabled devices is still at an early stage and faces significant challenges.

At present, most evidence originates from pilot studies or single-center research, often with small patient cohorts, which limits generalizability. Regulatory approval processes for AI/ML components are complex and evolving, leading to uncertainty in pathways for clinical adoption. Economic barriers, such as high device costs and limited reimbursement frameworks, further restrict widespread use, especially in smaller institutions and low-resource settings. In addition, clinician acceptance remains variable, with concerns about transparency, interpretability, medico-legal responsibility, and integration into existing clinical workflows. Another critical limitation is the need for continuous algorithm retraining and validation, as AI models can degrade in performance when exposed to new populations or changing practice environments. Federated learning and multicenter

collaborations may help overcome these barriers, but such strategies are still in early development. Data privacy and cybersecurity also remain pressing concerns for connected AI-enabled devices.

Despite these limitations, the trajectory of AI/ML in cardiology is promising. Continued advances in algorithm development, rigorous clinical validation, transparent regulatory frameworks, and health-economic evidence will be essential to translate early successes into widespread real-world adoption. Rather than replacing clinicians, AI/ML technologies should be seen as augmenting human expertise, enabling more informed, timely, and patient-centered cardiovascular care.

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